AWS SAA-C03 Exam Notes: Route 53

1. What is DNS? (Domain Name System)

- **Concept:** The "Phonebook of the Internet." It translates human-friendly hostnames (e.g., `www.google.com`) into machine-readable IP addresses (e.g., `172.217.1.1`).
- **Hierarchy:**
  - **Root:** The dot at the end (managed by IANA).
  - **TLD (Top Level Domain):** .com, .org, .gov.
  - **SLD (Second Level Domain):** amazon.com, https://www.google.com/search?q=google.com.
  - **Subdomain:** www, api, blog.
- **Records:** The specific instructions in the DNS database (A, AAAA, CNAME, NS, SOA).

2. Route 53 Overview

- **Definition:** A highly available and scalable cloud Domain Name System (DNS) web service.
- **Key Functions:**
  1. **Domain Registration:** You can buy domain names.
  2. **DNS Routing:** Routing internet traffic to resources.
  3. **Health Checking:** Monitoring the health of resources.
- **Global Service:** Route 53 is a **Global** service (similar to IAM and S3), though resources you point to are Regional.
- **SLA:** It is the only AWS service to offer a **100% Availability SLA**.
- **Hosted Zones:**
  - **Public Hosted Zone:** Records for the internet to resolve.
  - **Private Hosted Zone:** Records for internal VPCs to resolve (requires VPC settings `enableDnsHostnames` and `enableDnsSupport` to be true).

3. Route 53 - TTL (Time To Live)

- **Definition:** The duration (in seconds) that a DNS resolver or local computer caches a DNS record.
- **Mandatory:** TTL is mandatory for all record types **except Alias records**.
- **High TTL (e.g., 24 hours):**
  - Less traffic to Route 53.
  - Lower cost.
  - **Con:** Changes to records take a long time to propagate.
- **Low TTL (e.g., 60 seconds):**

- More traffic to Route 53.
- Higher cost.
- **Pro:** Changes propagate quickly (good for changing IPs frequently).

4. Route 53 CNAME vs Alias (CRITICAL EXAM TOPIC)

CNAME (Canonical Name)

- Points a **Hostname to another Hostname**.
- Example: `app.mydomain.com` -> `lb-123.us-east-1.elb.amazonaws.com`.
- **Limitation:** You **CANNOT** create a CNAME record for the **Zone Apex** (Root Domain/Naked Domain), e.g., `mydomain.com`.

Alias Record

- **AWS Specific** feature.
- Points a **Hostname to an AWS Resource** (ELB, CloudFront, S3 Website, Elastic Beanstalk, API Gateway, VPC Interface Endpoint, Global Accelerator).
- **Key Advantage:** It **CAN** be used at the **Zone Apex** (`mydomain.com`).
- **Behavior:** It acts like an A record (returns an IP) but internally maps to the AWS resource.
- **Cost:** Alias queries to AWS resources are free.
- **TTL:** You cannot set a TTL for Alias records (Route 53 respects the TTL of the target resource).

> **Exam Tip:** If a question asks to point the *root domain* (zone apex) to an ELB or S3 bucket, the answer is always **Alias**. Never CNAME.

5. Routing Policies

A. Simple Routing Policy

- **Use Case:** Use when you have a single resource that performs a given function for your domain.
- **Behavior:**
  - Can return one or multiple values (IPs) to the client.
  - If multiple values are returned, the client chooses one at random.
- **Constraint:** Does **not** support Health Checks (you cannot route away from an unhealthy instance automatically with Simple routing).

B. Weighted Routing Policy

- **Use Case: A/B Testing**, Blue/Green Deployments, or splitting traffic between regions intentionally.

- **Behavior:** You assign a "Weight" (0-255) to each record.
- **Logic:** Traffic % = (Specific Weight / Sum of all Weights).
  - *Example:* Record A (Weight 10), Record B (Weight 90). Record A gets 10% traffic.
- **Note:** Setting weight to 0 stops traffic to that resource.

C. Latency Routing Policy

- **Use Case: Performance**. You want to provide the best experience for global users.
- **Behavior:** Route 53 looks at the user's location and the AWS Region latency, then routes the user to the AWS Region with the **lowest latency** (fastest response time).

D. Failover Routing Policy

- **Use Case: Disaster Recovery (DR)**. Active-Passive configuration.
- **Prerequisite:** Requires **Health Checks**.
- **Behavior:**
  - **Primary Record:** Route 53 monitors this.
  - **Secondary Record:** Traffic is routed here *only* if the Primary is unhealthy.

E. Geolocation Routing Policy

- **Use Case: Compliance**, Content Restrictions, Localized Language.
- **Behavior:** Routes traffic based on the **geographic location of the user** (Continent, Country, or US State).
- **Example:** Users in France go to French servers; everyone else goes to a Default server.
- **Key Difference from Latency:** Latency is about *speed*; Geolocation is about *boundaries*.

F. Geoproximity Routing Policy

- **Use Case:** Complicated geographic routing with "Bias."
- **Behavior:** Routes traffic based on the geographic location of the users and the resources.
- **Bias:** You can shift traffic from one region to another by increasing/decreasing the "bias" (expand or shrink the geographic region coverage).
- **Requirement:** Must use **Route 53 Traffic Flow** (visual editor).

G. IP-Based Routing Policy

- **Use Case:** Performance optimization for specific ISPs or routing internal subnets.
- **Behavior:** Routes traffic based on the client's **CIDR (IP)** block.
- **Example:** Route 53 recognizes the IP is from "Comcast" or a specific corporate range and routes accordingly.

H. Multi-Value Answer Routing Policy

- **Use Case:** "Simple Routing with Health Checks."
- **Behavior:** Returns multiple values (IPs) in response to DNS queries, but **only returns values for resources that are Healthy**.
- **Limit:** Up to 8 records.
- **Note:** It is not a substitute for an ELB (Load Balancer), but acts as a basic client-side load balancer.

6. Health Checks

- **Purpose:** Automate failover and monitor status.
- **Types:**
  1. **Endpoint:** Checks an IP address or domain name (HTTP/HTTPS/TCP).
  2. **Calculated:** Combines results of other health checks (AND/OR/NOT logic).
  3. **CloudWatch Alarm:** Checks the state of a CloudWatch Alarm (useful for checking DynamoDB throttle or custom metrics, not just connectivity).
- **Operation:**
  - Global health checkers send requests to your endpoint.
  - Default: 30-second interval (can be fast - 10s).
  - Default: 3 retries constitute a failure.
- **Private Hosted Zones:** If monitoring an internal resource (Private IP), you must create a CloudWatch Alarm and use the "CloudWatch Alarm" health check type (Route 53 public checkers cannot access your private VPC).

7. 3rd Party Domains & Route 53

- **Scenario:** You bought `example.com` on GoDaddy (Registrar) but want to manage records in AWS Route 53.
- **Steps:**
  1. Create a **Hosted Zone** in Route 53 for `example.com`.
  2. Route 53 generates 4 **Name Servers (NS)**.
  3. Log in to GoDaddy (Registrar).
  4. Update the NS records in GoDaddy to match the 4 NS records provided by Route 53.
- **Note:** Domain Registration and DNS Hosting are separate. You can keep the registration at GoDaddy but host the DNS zone file at AWS.

8. Route 53 Resolvers & Hybrid DNS

- **Scenario:** You have an On-Premise Data Center and an AWS VPC connected via Direct Connect or VPN. You want on-prem servers to resolve AWS names, and AWS instances to resolve on-prem names.
- **Route 53 Resolver (Inbound Endpoint):**

    - Allows **On-Premise** resources to query Route 53 Resolver.
    - Forward DNS queries from On-Prem -> AWS.
- **Route 53 Resolver (Outbound Endpoint):**
    - Allows **AWS** resources to query On-Premise DNS servers.
    - Forward DNS queries from AWS -> On-Prem (using Forwarding Rules).

Summary of Exam "Gotchas"

Keyword/Scenario	Answer usually involves...
Zone Apex / Root Domain	**Alias Record** (Not CNAME)
Active/Passive DR	**Failover Routing**
Lowest Latency / Fastest	**Latency Routing**
Legal Compliance / Borders	**Geolocation Routing**
Traffic Shift / Bias	**Geoproximity Routing** (Traffic Flow)
80% to V1, 20% to V2	**Weighted Routing**
Hybrid Cloud DNS	**Route 53 Resolver** (Inbound/Outbound)
Health Check Private IP	**CloudWatch Alarm** Health Check